



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2012
YEAR 11 Mathematics Extension
HSC Task #1

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answers must be given in simplest exact form unless otherwise stated.

Total marks - 77

Section I (10 marks)

- Answer Questions 1-10 on the Multiple Choice answer sheet provided.

Section II (67 marks)

- For Questions 11-13, start a new answer booklet for each question.

Examiner: *D.McQuillan*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

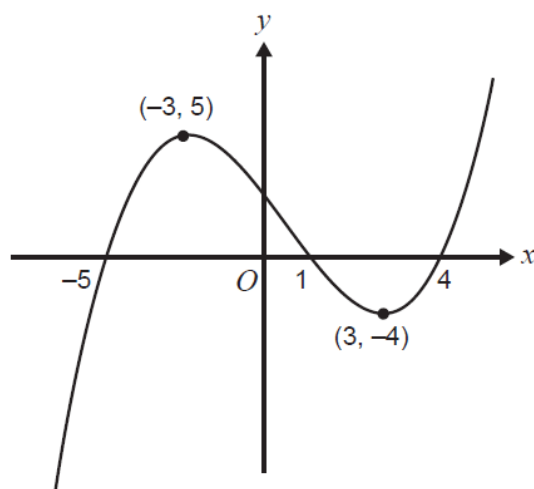
Section I (10 marks)

Use the multiple-choice answer sheet for Questions 1-10

- 1 What is the acute angle to the nearest degree between the lines $y = 7 - 4x$ and $2x - 3y - 6 = 0$?
- (A) 26°
- (B) 42°
- (C) 70°
- (D) 75°
- 2 When polynomial $P(x)$ is divided by $x^2 + x - 6$ the remainder is $7x + 13$. What is the remainder when $P(x)$ is divided by $x + 3$?
- (A) -8
- (B) -5
- (C) 34
- (D) 55
- 3 What is the exact value of $\cos 165^\circ$?
- (A) $\frac{-\sqrt{2}-\sqrt{6}}{4}$
- (B) $\frac{-\sqrt{2}+\sqrt{6}}{4}$
- (C) $\frac{-\sqrt{6}-\sqrt{2}}{4}$
- (D) $\frac{-\sqrt{6}+\sqrt{2}}{4}$

- 4 What are the coordinates of the point P that divides internally the interval joining the points A(1, 2) and B(7, 5) in the ratio 2:1 ?
- (A) (3, 3)
- (B) (3, 4)
- (C) (5, 3)
- (D) (5, 4)
- 5 What is the solution to the inequality $\frac{3}{x-2} \geq 4$?
- (A) $x < -2$ and $x \geq \frac{11}{4}$
- (B) $x > -2$ and $x \leq -\frac{11}{4}$
- (C) $x < 2$ and $x \geq \frac{11}{4}$
- (D) $x > 2$ and $x \leq \frac{11}{4}$
- 6 The average rate of change of the function $f(x) = x^3 - \sqrt{x+1}$ between $x = 0$ and $x = 3$ is
- (A) 12
- (B) $\frac{26}{3}$
- (C) $\frac{25}{3}$
- (D) 8

7



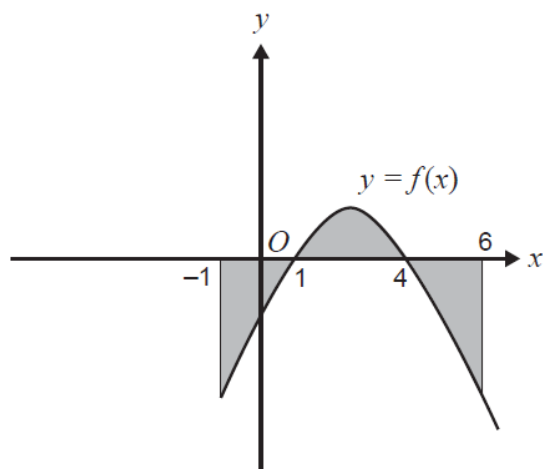
For the graph of $y = f(x)$ shown above $f'(x)$ is negative when

- (A) $-3 < x < 3$
- (B) $-3 \leq x \leq 3$
- (C) $x < -3$ or $x > 3$
- (D) $x < 5$ or $1 < x < 4$

8 Let $f'(x) = g'(x) + 3$, $f(0) = 2$ and $g(0) = 1$.

Then $f(x)$ is given by

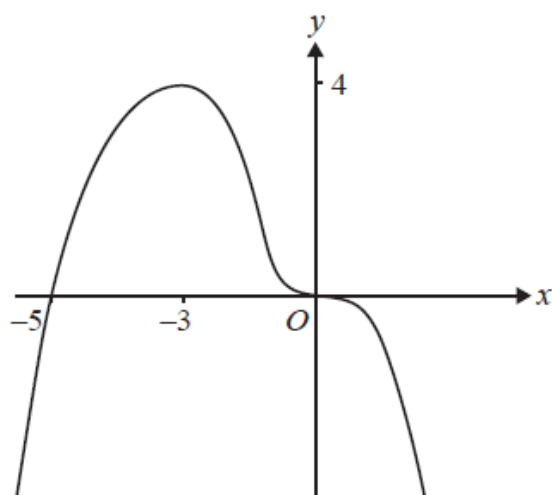
- (A) $f(x) = g(x) + 3x + 1$
- (B) $f(x) = g(x) + 3x$
- (C) $f(x) = 1$
- (D) $f(x) = g'(x) + 3$



The total area of the shaded regions in the diagram is given by

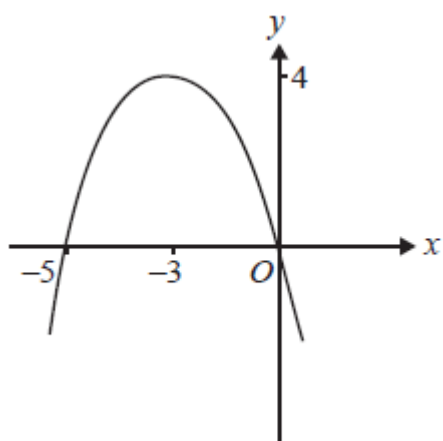
- (A) $\int_{-1}^6 f(x) dx$
- (B) $\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx + \int_4^6 f(x) dx$
- (C) $-\int_1^{-1} f(x) dx + \int_1^4 f(x) dx - \int_6^4 f(x) dx$
- (D) $\int_1^{-1} f(x) dx + \int_1^4 f(x) dx + \int_6^4 f(x) dx$

- 10 The graph of $y = f(x)$ is show below.

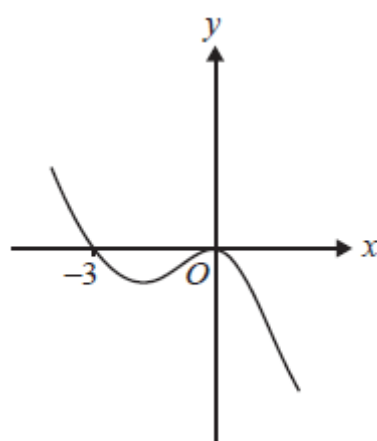


Which of the following could be the graph of the derivative function $y = f'(x)$?

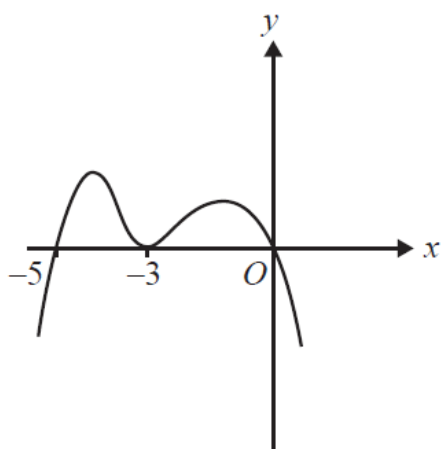
(A)



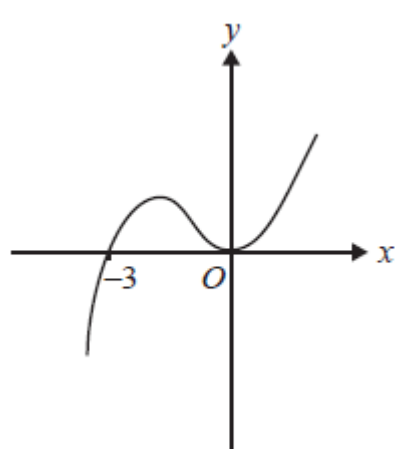
(B)



(C)



(D)



Section II

Start a new writing booklet for each Question in section II

Question 11 (21 marks)

(a) Find [4]

(i)

$$\int (5 - 2x + 6x^2) dx$$

(ii)

$$\int \sqrt[3]{x} \, dx$$

(iii)

$$\int \frac{5x^4 + x - 4}{2x^6} dx$$

(b) Solve [4]

(i)

$$\frac{4}{|x - 1|} > 1$$

(ii)

$$\frac{4}{(x - 1)^2} > 1$$

(c) Given $P(x) = x^3 - x^2 - 3x - 1$ [4]

(i) Evaluate $P(-1)$.

(ii) Hence solve $P(x) = 0$.

(d) The roots of $x^3 - 4x^2 - 9 = 0$ are α, β, γ . Find the value of [4]

(i) $(\alpha + \beta + \gamma) - \alpha\beta\gamma$

(ii) $(\alpha + 2)(\beta + 2)(\gamma + 2)$

(e) Prove that [2]

$$\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A$$

(f) Sketch the graph of a function that satisfies all of the given conditions. [3]

- $f'(5) = 0$
- $\lim_{x \rightarrow 3} f(x) = -\infty$
- $f'(x) < 0$ for $x < 3$ and for $x > 5$
- $f'(x) > 0$ for $3 < x < 5$

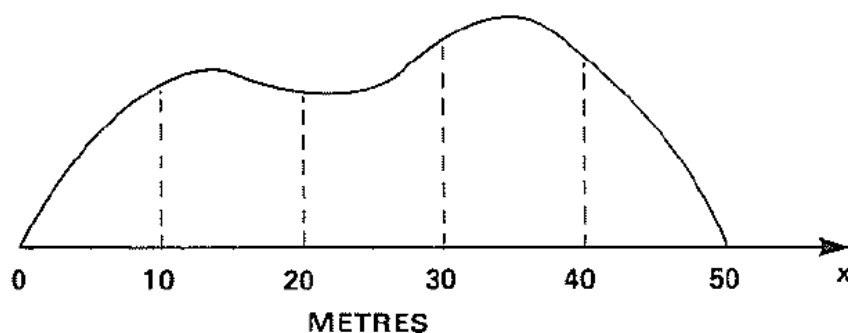
End of Question 11

Start a new writing booklet for each Question in section II

Question 12 (23 marks)

- (a) During a survey the area of an irregular headland was to be found. The surveyor used a base line divided into five equal sub-intervals, each of width ten metres. Offset measurements $f(x)$ were taken at each value x across the base line and tabulated as below

[3]



x (metres)	0	10	20	30	40	50
Offset (metres)	0	13.2	13.8	17	17.4	0

Use the trapezoidal rule to find the approximate area of the headland.

(b)

[4]

(i) Rewrite $2 \cos x + \sqrt{5} \sin x$ in the form $R \cos(x - \alpha)$.

(ii) Hence solve $2 \cos x + \sqrt{5} \sin x = 2$ for $0^\circ \leq x < 360^\circ$, to the nearest degree.

(c) Sketch $y = x^4 - 6x^2 - 8x + 1$ showing all stationary points.

[5]

(d) If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ possibly be?

[2]

(e) Find the general solution of $\cos 2x = \frac{1}{2}$ (answer in terms of radians). [2]

(f) [4]

(i) Show that

$$\frac{d}{dx} \left(\frac{x}{a^2 \sqrt{a^2 - x^2}} \right) = \frac{1}{\sqrt{(a^2 - x^2)^3}}$$

(ii) Hence evaluate

$$\int_0^2 \frac{dx}{\sqrt{(9 - x^2)^3}}$$

(g) Prove that the polynomial

$$P(x) = x^{101} + x^{51} + x + 1$$

has neither a local maximum nor a local minimum. [3]

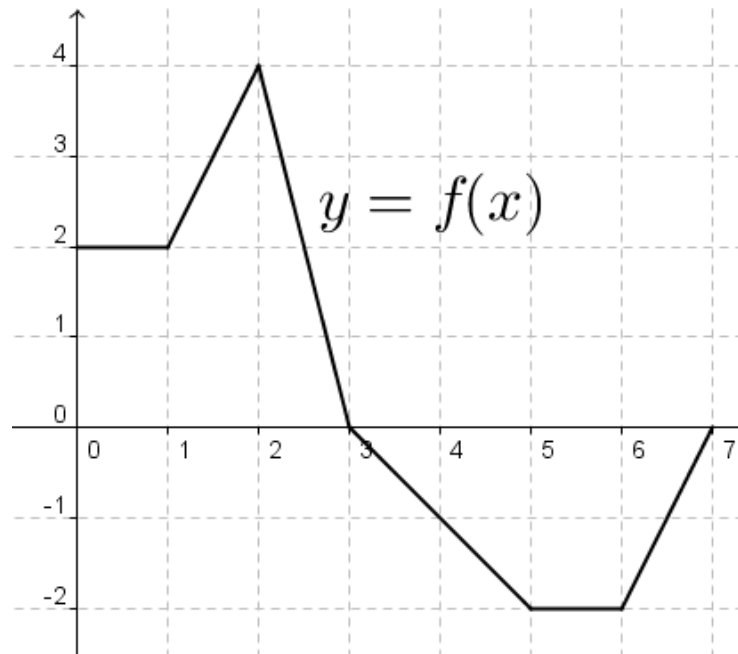
End of Question 12

Start a new writing booklet for each Question in section II

Question 13 (23marks)

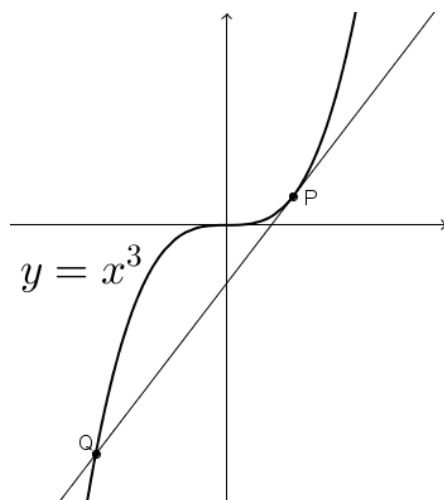
(a) Let $g(x) = \int_0^x f(t)dt$ where $f(x)$ is the function whose graph is shown.

[6]



- (i) Evaluate $g(1)$, $g(3)$ and $g(6)$.
- (ii) On what intervals is $g(x)$ increasing?
- (iii) Where does $g(x)$ have a maximum value?
- (iv) Sketch the graph of $y = g(x)$.

- (b) Let P be a point on the curve $y = x^3$ and suppose the tangent line at P intersects the curve again at Q. Prove that the slope at Q is four times the slope at P. [4]



- (c) A square room with length and breadth 4 metres is 2.4 metres high. Calculate, to the nearest minute, the angle which a string line will make with the floor if it is stretched from the centre of the ceiling to one corner of the floor. [3]

- (d) Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side L if one side of the rectangle lies on the base of the triangle. [4]

- (e) If $4 \sin 2x + 2 \cos 2x = \tan x$ [6]

(i) Prove that $\tan x$ is a root of the equation $t^3 + 2t^2 - 7t - 2 = 0$.

(ii) Verify that $t = 2$ is one root and find the exact values of the others.

(iii) Hence solve, $4 \sin 2x + 2 \cos 2x = \tan x$ for $0^\circ < x < 360^\circ$, to the nearest degree.

End of Question 13

End of Exam



Student Number: _____

SOLNS

Mathematics Extension Task 1 2012

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. A B ☒ C D
2. ☒ A B C D
3. ☒ A B ☒ C D
4. A B C ☒ D
5. A B C ☒ D
6. A ☒ B C D
7. ☒ A B C D
8. ☒ A B C D
9. A B C ☒ D
10. A ☒ B C D

C
A
A or C
D
D
B
A
A
D
B

Q11 (EXT1)

(a) (i) $\int (5 - 2x + 6x^2) dx = 5x - x^2 + 2x^3 + C.$

[1]

(ii) $\int \sqrt[3]{x} dx = \int x^{\frac{1}{3}} dx.$

$$= \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + C.$$

$$= \frac{3}{4} x^{\frac{4}{3}} + C \quad \text{OR} \quad \frac{3x\sqrt[3]{x}}{4} + C.$$

[1]

(iii) $\int \frac{5x^4 + x - 4}{2x^6} dx = \int \left(\frac{5}{2} x^{-2} + \frac{x^{-5}}{2} - 2x^{-6} \right) dx$

$$= \frac{5x^{-1}}{-2} + \frac{x^{-4}}{-8} - \frac{2x^{-5}}{-5} + C$$

$$= \left[-\frac{5}{2} x^{-1} - \frac{1}{8} x^{-4} + \frac{2}{5} x^{-5} + C \right]$$

$$\text{OR} = \frac{-5}{2x} - \frac{1}{8x^4} + \frac{2}{5x^5} + C$$

$$\text{OR} \quad \left[\frac{-100x^4 - 5x + 16}{40x^5} + C \right]$$

[2]

(b) (i) $\frac{4}{|x-1|} > 1$

$$\Rightarrow |x-1| < 4$$

$$\text{ie } -4 < x-1 < 4$$

$$\text{ie } \boxed{-3 < x < 5} \quad \boxed{x \neq 1}$$

[2]

(ii) $\frac{4}{(x-1)^2} > 1$
 $4 > (x-1)^2$

$$-2 < x-1 < 2$$

$$\boxed{-1 < x < 3, x \neq 1}$$

[2]

$$(c) \quad P(x) = x^3 - x^2 - 3x - 1.$$

$$(i) \quad P(-1) = -1 - 1 + 3 - 1 \\ = 0.$$

$\therefore x+1$ is a factor.

$$-1 \left[\begin{array}{cccc} 1 & -1 & -3 & -1 \\ & -1 & 2 & 1 \\ \hline & 1 & -2 & -1 & 0 \end{array} \right] \quad \left[\begin{array}{l} \text{using Synthetic} \\ \text{or Long Division} \end{array} \right]$$

$\therefore x^2 - 2x - 1$ is the other factor.

$$\therefore P(x) = (x+1)(x^2 - 2x - 1)$$

$$\text{If } P(x) = 0$$

$$(x+1)(x^2 - 2x - 1) = 0$$

$$x = -1, \frac{2 \pm \sqrt{8}}{2}$$

$$\boxed{x = -1; 1 \pm \sqrt{2}.}$$

$$(d) \quad \text{given } x^3 - 4x^2 - 9 = 0 \text{ with roots } \alpha, \beta, \gamma.$$

$$\alpha + \beta + \gamma = 4$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0$$

$$\alpha\beta\gamma = 9.$$

$$\therefore (i) \quad (\alpha + \beta + \gamma) - \alpha\beta\gamma = 4 - 9 \\ \boxed{= -5}$$

$$(ii) \quad (\alpha+2)(\beta+2)(\gamma+2) = \alpha\beta\gamma + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ + 4(\alpha + \beta + \gamma) + 8. \\ = 9 + 0 + 16 + 8 \\ = \boxed{33}$$

(e) Prove that $\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A$.

$$\text{LHS} = \frac{1 - (1 - 2\sin^2 A)}{1 + (2\cos^2 A - 1)}$$

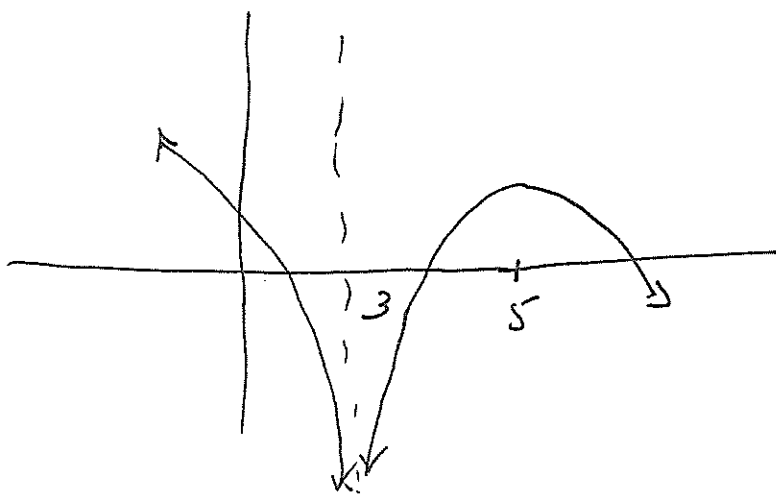
[2]

$$= \frac{2\sin^2 A}{2\cos^2 A}$$

$$= \tan^2 A$$

$$= \text{RHS.}$$

(f)



[3]

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Question 12

$$\begin{aligned}
 (a) A &\doteq \frac{h}{2} [y_0 + y_5 + 2(y_1 + y_2 + y_3 + y_4)] \\
 &= 5 [0 + 0 + 2(13.2 + 13.8 + 17 + 17.4)] \\
 &= 614 \text{ m}^2 \quad [3]
 \end{aligned}$$

$$\begin{aligned}
 (b)(i) 2\cos x + \sqrt{5}\sin x &= R \cos(x-a) \\
 R &= \sqrt{2^2 + 5^2} \\
 &= 3
 \end{aligned}$$

$$\therefore \text{LHS} = 3 \left(\frac{2}{3} \cos x + \frac{\sqrt{5}}{3} \sin x \right)$$

$$\therefore \sin a = \frac{\sqrt{5}}{3}$$

$$\begin{aligned}
 a &\doteq 48^\circ 11' 23'' \\
 &\doteq 0.8411^\circ \quad [2]
 \end{aligned}$$

$$\begin{aligned}
 (ii) E_{\text{op}} &= 3 \cos(x - 0.8411) \\
 &= 3 \cos(x - 48^\circ 11' 23'')
 \end{aligned}$$

$$3 \cos(x - 48^\circ 11' 23'') = 2$$

$$x - 48^\circ 11' 23'' = \cos^{-1}\left(\frac{2}{3}\right)$$

$$= -48^\circ 11' 23'' \text{ or } 48^\circ 11' 23''$$

$$x = 0^\circ \text{ or } 96^\circ 22' 46''$$

$$\doteq 0^\circ \text{ or } 96^\circ \quad [2]$$

$$(c) y = x^4 - 6x^2 - 8x + 1$$

$$y' = 4x^3 - 12x - 8$$

$$y'' = 12x^2 - 12$$

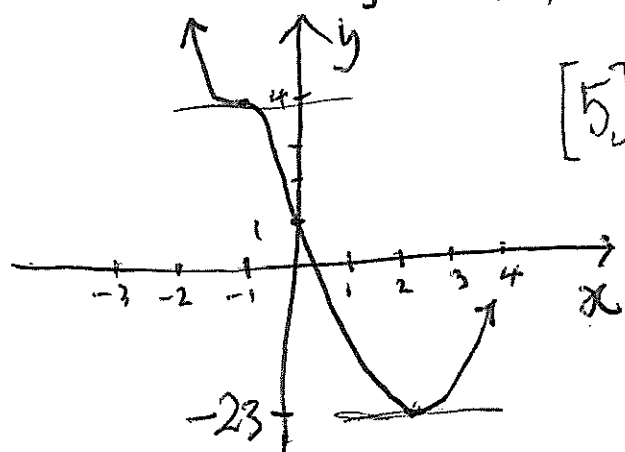
$$y' = 0 \text{ when}$$

$$4x^3 - 12x - 8 = 0$$

$$x^3 - 3x - 2 = 0$$

$$x = -1, -1, 2$$

$$y = 4, -23$$



(d) Min of $f(4)$ will be when $y'(x) = 2$ (least value)

$\therefore$ let $m=2, (1, 10)$

$$y - 10 = 2(x - 1)$$

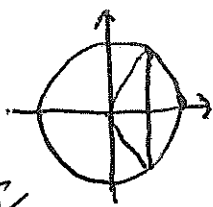
$$y = 2x + 8$$

When $x=4$

$$y = 8 + 8 = 16$$

$$\therefore f(4) \geq 16 \quad [2]$$

$$(e) \cos 2x = \frac{1}{2}$$



$$2x = 2n\pi \pm \frac{\pi}{3}$$

$$x = n\pi \pm \frac{\pi}{6} \quad [2] \\ n \in \mathbb{Z}$$

$$(f) (i) \frac{d}{dx} \left(\frac{x}{a^2 \sqrt{a^2 - x^2}} \right)$$

$$= \frac{1}{a^2} \left[\frac{\sqrt{a^2 - x^2} \cdot 1 - x \cdot \frac{1}{2\sqrt{a^2 - x^2}} \cdot (-2x)}{(a^2 - x^2)} \right]$$

$$= \frac{1}{a^2} \left[\frac{(a^2 - x^2) + x^2}{\sqrt{(a^2 - x^2)^3}} \right]$$

$$= \frac{1}{\sqrt{(a^2 - x^2)^3}} \quad [2] \\ \text{as required.}$$

$$(ii) \int_0^2 \frac{dx}{\sqrt{(9 - x^2)^3}}$$

$$= \left[\frac{1}{9} \frac{x}{\sqrt{9 - x^2}} \right]_0^2$$

$$= \frac{1}{9} \left[\frac{2}{\sqrt{5}} - 0 \right]$$

$$= \frac{2}{9\sqrt{5}} \quad [2]$$

$$(g) P(x) = x^{101} + x^{51} + x + 1$$

$$P'(x) = 101x^{100} + 51x^{50} + 1$$

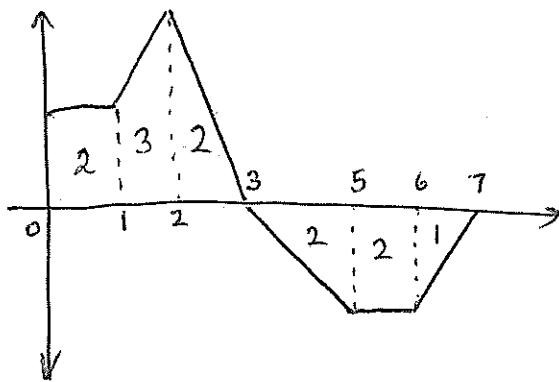
$$> 0 \quad \forall x$$

$\therefore P(x)$ is monotonic increasing.

$\therefore P(x)$ has no turning points. [3]

Question 13

a)



i) $g(1) = 2$

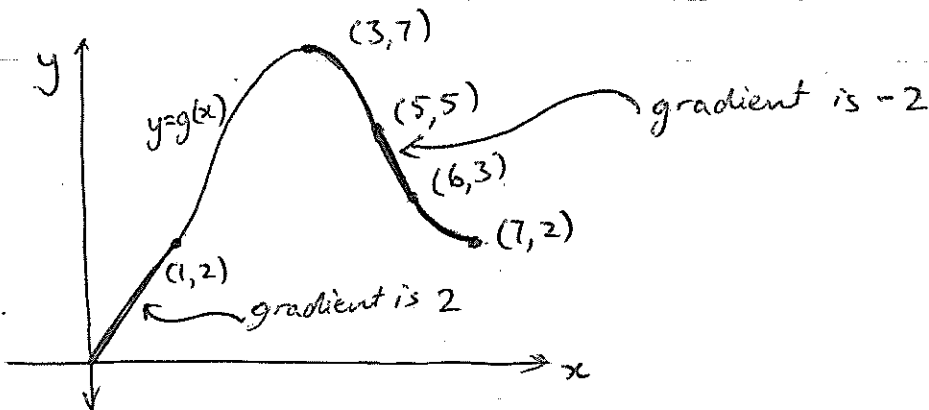
$g(3) = 7$

$g(6) = 3$

ii) $0 < x < 3$

iii) when $x = 3$

iv)



b) let P have coordinates (x_1, y_1)

Q have coordinates (x_2, y_2)

$$y = x^3$$

$$y' = 3x^2$$

at P

$$m_T = 3x_1^2$$

equation of tangent at P : $y - y_1 = 3x_1^2(x - x_1)$

$$y = 3x_1^2x - 3x_1^3 + y_1$$

$$y = 3x_1^2x - 2x_1^3 \quad \text{①, since } y_1 = x_1^3$$

$$y = x^3 \quad \text{②}$$

sub ① into ②

$$x^3 = 3x_1^2x - 2x_1^3$$

$$x^3 - 3x_1^2x + 2x_1^3 = 0$$

let roots be x_1, x_1, x_2 (since tangent at $P(x_1, y_1)$)

$$x_1 + x_1 + x_2 = 0$$

$$2x_1 + x_2 = 0$$

$$x_2 = -2x_1$$

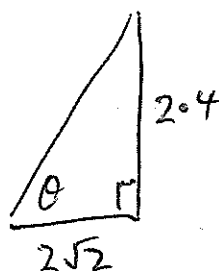
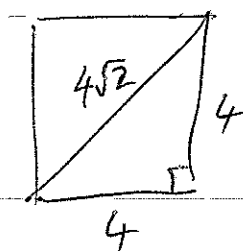
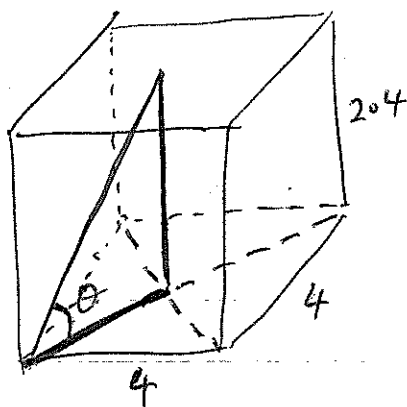
$$\text{gradient of tangent at } Q = 3(-2x_1)^2$$

$$= 12x_1^2$$

$$= 4(3x_1^2)$$

= four times gradient of tangent at P

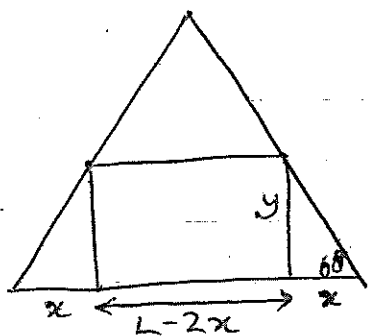
c)



$$\tan \theta = \frac{2 \cdot 4}{2\sqrt{2}}$$

$$\theta = 40^\circ 19'$$

d)



$$A = (L - 2x)y$$

$$\tan 60^\circ = \frac{y}{x}$$

$$\sqrt{3} = \frac{y}{x}$$

$$y = \sqrt{3}x$$

$$A = (L - 2x)\sqrt{3}x$$

$$A = \sqrt{3}Lx - 2\sqrt{3}x^2$$

$$A' = \sqrt{3}L - 4\sqrt{3}x$$

$$A'' = -4\sqrt{3}$$

$$\text{let } A' = 0$$

$$\sqrt{3}L - 4\sqrt{3}x = 0$$

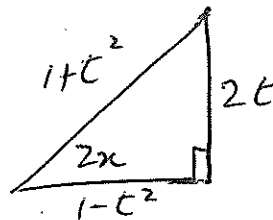
$$4\sqrt{3}x = \sqrt{3}L$$

$$x = \frac{L}{4}$$

since $A'' < 0$, Maximum area when $x = \frac{L}{4}$

Dimensions are $\frac{L}{2} \times \frac{\sqrt{3}L}{4}$

e) i) let $t = \tan x$



$$4 \sin 2x + 2 \cos 2x = \tan x$$

$$4 \left(\frac{2t}{1+t^2} \right) + 2 \left(\frac{1-t^2}{1+t^2} \right) = t$$

$$8t + 2 - 2t^2 = t + t^3$$

$$t^3 + 2t^2 - 7t - 2 = 0$$

ii) let $P(t) = t^3 + 2t^2 - 7t - 2$

$$P(2) = (2)^3 + 2(2)^2 - 7(2) - 2 = 0$$

$\therefore t = 2$ is a root of $P(t) = 0$.

$$\begin{array}{r} t^2 + 4t + 1 \\ t-2 \overline{) t^3 + 2t^2 - 7t - 2} \\ \underline{t^3 - 2t^2} \\ 4t^2 - 7t \\ \underline{4t^2 - 8t} \\ t - 2 \\ \underline{t - 2} \\ 0 \end{array}$$

$$\therefore P(t) = (t-2)(t^2 + 4t + 1)$$

$$t^2 + 4t + 1 = 0$$

$$t^2 + 4t + 4 = -1 + 4$$

$$(t+2)^2 = 3$$

$$t+2 = \pm\sqrt{3}$$

$t = -2 \pm \sqrt{3}$ are the other roots

$$\text{iii)} \quad \tan x = 2$$

$$\tan \alpha = 2$$

$$\alpha \doteq 63^\circ$$

$$x = 63^\circ, 243^\circ$$

S	A ✓
✓ T	C

$$\tan x = -2 + \sqrt{3}$$

$$\tan \alpha = 2 - \sqrt{3}$$

$$\alpha = 15^\circ$$

$$x = 165^\circ, 345^\circ$$

✓ S	A
T	✓ C

$$\tan x = -2 - \sqrt{3}$$

$$\tan \alpha = 2 + \sqrt{3}$$

$$\alpha = 75^\circ$$

$$x = 105^\circ, 285^\circ$$

✓ S	A
T	✓ C

$$\therefore x = 63^\circ, 105^\circ, 165^\circ, 243^\circ, 285^\circ, 345^\circ$$